

Milestone-3 Report

ExoHabitAI – Backend & Frontend Integration

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Project: ExoHabitAI

Module: 5 & 6 (Backend & Frontend)

1. Introduction

This milestone focuses on integrating the trained machine learning model with a Flask backend API and developing a frontend interface for user interaction.

2. Backend Implementation

The backend was developed using Flask. It loads the trained Random Forest model and provides REST API endpoints.

Key Features:

- Flask framework
- Model loading using joblib
- /predict endpoint
- JSON input and output
- Error handling
- CORS enabled

The backend receives planetary parameters and returns habitability prediction and probability.

3. Backend Workflow

1. User sends input from frontend
2. Request reaches Flask API
3. Model processes data
4. Prediction is generated
5. Response is sent back

This ensures smooth integration between ML and UI.

4. Frontend Implementation

The frontend was built using HTML, CSS, and JavaScript.

Components:

- Input form
- Predict button
- Result display
- Error alerts

The interface is clean and user-friendly.

5. Frontend–Backend Integration

The frontend uses Fetch API to communicate with Flask backend.

Features:

- Sends POST request
- JSON format
- Displays response
- Handles errors

This allows real-time prediction.

6. Testing and Validation

The system was tested using:

- Browser interface
- Curl commands
- Sample inputs

Correct predictions were received.

7. Results

The integrated system successfully predicts whether an exoplanet is habitable.

Outputs include:

- Status
- Probability score

The interface displays results clearly.

8. Conclusion

This milestone successfully implemented backend APIs and frontend integration. The system is functional, responsive, and ready for deployment.

9. Screenshots

The screenshot shows a web browser window with the address bar displaying the file path: `/Users/amruthavarshini/Desktop/ExoHabitAI/ExoHabitAI/frontend/index.html`. The browser's address bar also shows navigation icons (back, forward, refresh) and a star icon for bookmarks. The main content area of the browser displays the "ExoHabitAI Habitability Predictor" interface. The interface has a light purple background. At the top, the title "ExoHabitAI Habitability Predictor" is centered. Below the title, there is a white rounded rectangle containing three input fields. The first input field is labeled "Planet Temperature (pl_eqt)" and contains the value "280". The second input field is labeled "Planet Radius (pl_rade)" and contains the value "1". The third input field is labeled "Orbital Distance (pl_orbsmax)" and contains the value "1". Below these input fields is a blue button labeled "Predict". Below the white rounded rectangle, there is a green rounded rectangle containing the text "Status: Habitable 🌍" and "Probability: 1".